

Programming assignment 10

1) 2-3 trees

1a) Y L P M X H C R A E S T B C A

1. Y

Y

2. L

LY

3. P (shifts)

LPY →
L PY
L PY

4. M

P
LM Y

5. X

P
LM XY

6. H (shifts)

P P
HLM XY →
H LP XY
H M XY

7. C

LP
CH M XY

8. R (2 shifts)

LP
ch M RXY
→ LPX
ch M RY
→ P
ch M RY

9. A (shifts)

P
L X
ACH M RY
→ P
CL X
A H M RY

10. E

P
CL X
A E H M RY

11. S

P
CL X
A E H M R S Y

12. T (shifts)

P
CL X
A E H M R S T Y
→ P
CL SX
A E H M R T Y

13. B

P
CL SX
AB E H M R T Y

14. C

repeating elements,
so no change
in structure

15. A

1b) A L G O R I T H M S X Y Z

1. A

A

2. L

AL

3. G

AGL
→ G
A L

4. O

G
A LO

5. R

G
A LOR

6. I

GO
A IL R

7. T

GO
A IL RT

8. H

GO
A HIL RT
→ GIO
A H L RT
→ I
A H L RT

9. M

I
G O
A H LM RT

10. S

I
G O
A H LM RST
→ I
G OS
A H LM RT

11. X

I
G OS
A H LM R TX

12. Y

I
G OS
A H LM R TXY
→ I
G OSX
A H LM R TY

13. Z

IS
G O X
A H LM R TY

13. Z

IS
G O X
A H LM R T Y Z

2) Red black trees

2a) Y L P M X H C R A E S T B C A

1. Y Y

2. L L Y

3. P Y Y rotate L left
L P Y rotate Y right
L P Y color flip P
L P Y

4. M L P Y rotate L left
L M P Y

5. X L M P Y X

6. H L P Y rotate M right, color flip L
H L M X P Y
H L M X

7. C L P Y
H L M X C

8. R L P Y rotate Y right
C H L M X R color flip X, P
C H L M R Y

9. A L P X rotate H right, color flip C
A C H L M R Y
A C H L M R Y

10. E L P X
C L M R Y
A H E

11. S L P X rotate R left
A C H E L M R S Y
A C H E L M R S Y

12. T L P X color flip S
A C H E L M R T S Y
A C H E L M R T S Y

13. B L P X rotate A left
A C H B E L M R T S Y
A C H B E L M R T S Y

14. C

15. A

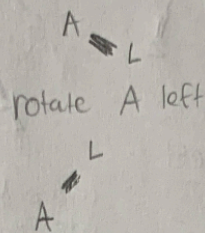
No change in structure

2b) A L G O R I T H M S X Y Z J D

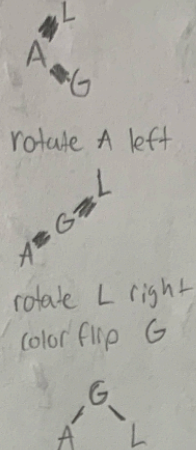
1. A

A

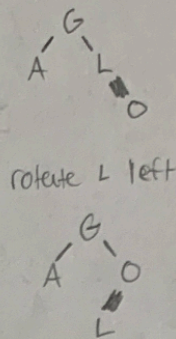
2. L



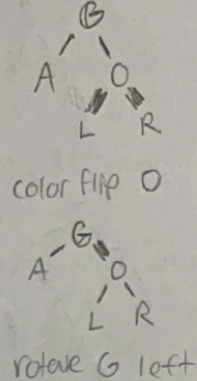
3. G



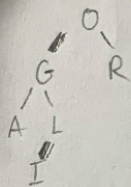
4. O



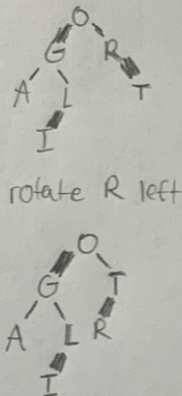
5. R



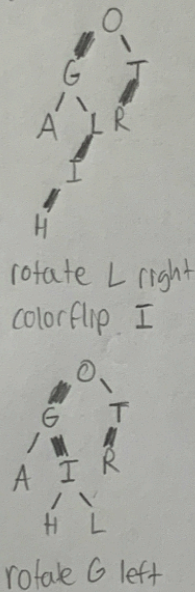
6. I



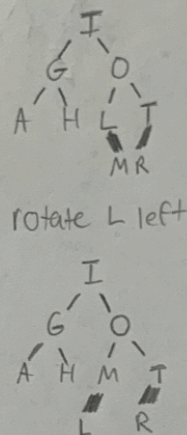
7. T



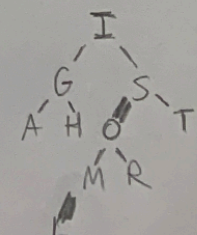
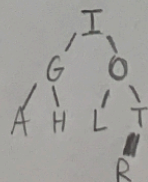
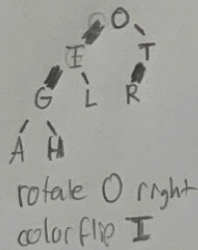
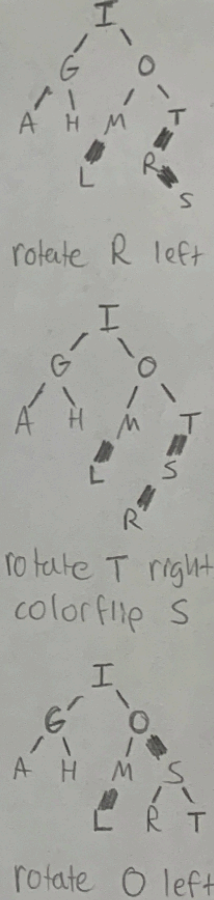
8. H

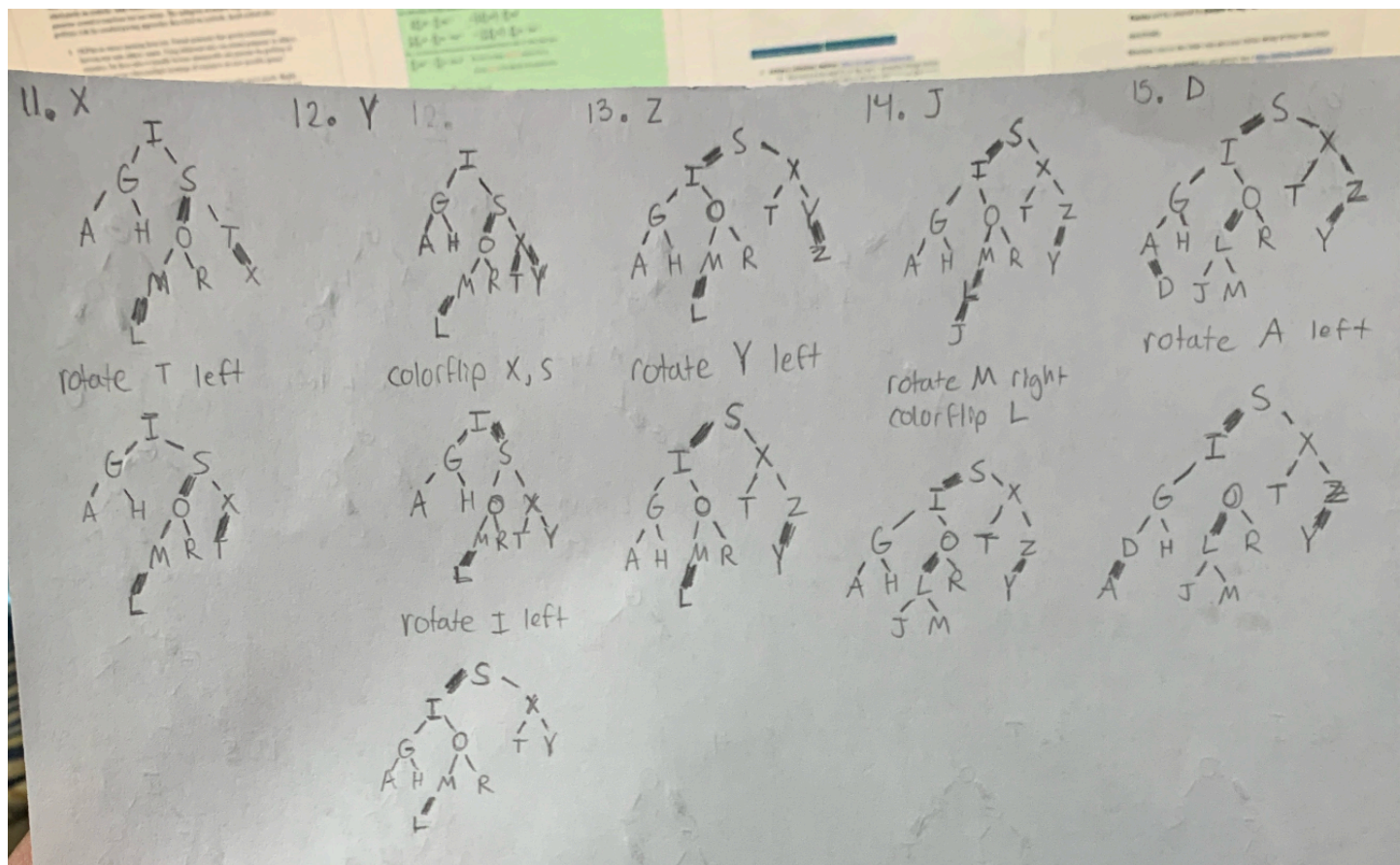


9. M

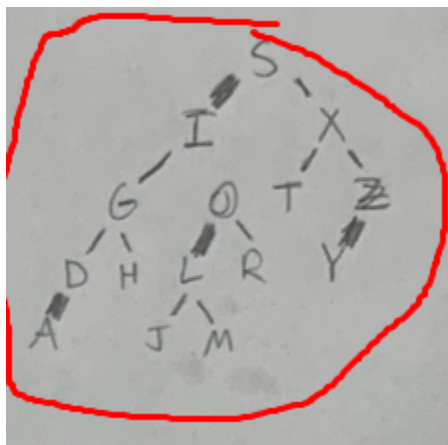


10. S





Note to self, answer is bottom right:



Programming output:

